

Derivative rules

- $\frac{d}{dx}[c \cdot f(x)] = c \cdot f'(x)$
- $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$
- $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$
- $\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$
- $\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$

Derivatives

- $\frac{d}{dx}[c] = 0$
- $\frac{d}{dx}[x^p] = px^{p-1}$ if $p \neq 0$
- $\frac{d}{dx}[\sin(x)] = \cos(x)$
- $\frac{d}{dx}[\cos(x)] = -\sin(x)$
- $\frac{d}{dx}[\tan(x)] = \sec^2(x)$
- $\frac{d}{dx}[\csc(x)] = -\csc(x) \cot(x)$
- $\frac{d}{dx}[\sec(x)] = \sec(x) \tan(x)$
- $\frac{d}{dx}[\cot(x)] = -\csc^2(x)$
- $\frac{d}{dx}[b^x] = b^x \ln(b)$
- $\frac{d}{dx}[e^x] = e^x$
- $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$

Vectors

- When working in $\mathbb{R}^3$, $\vec{i} = \langle 1, 0, 0 \rangle$, $\vec{j} = \langle 0, 1, 0 \rangle$, and $\vec{k} = \langle 0, 0, 1 \rangle$
- $||\langle a, b, c \rangle|| = \sqrt{a^2 + b^2 + c^2}$
- $\langle x_1, y_1, z_1 \rangle \bullet \langle x_2, y_2, z_2 \rangle = x_1x_2 + y_1y_2 + z_1z_2$
- $\vec{a} \bullet \vec{b} = ||\vec{a}|| \cdot ||\vec{b}|| \cos(\theta)$
- $\langle x_1, y_1, z_1 \rangle \times \langle x_2, y_2, z_2 \rangle = \det \begin{bmatrix} \vec{i} & \vec{j} & \vec{k} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{bmatrix}$
- $||\vec{a} \times \vec{b}|| = ||\vec{a}|| \cdot ||\vec{b}|| \sin(\theta)$

Multivariable derivatives

- $\nabla f(x, y) = \langle f_x, f_y \rangle$
- $D_{\vec{u}}f(x, y) = (\nabla f(x, y)) \bullet \vec{u}$
- $D(x, y) = f_{xx}(x, y)f_{yy}(x, y) - f_{xy}(x, y)^2$
- $\nabla f(x, y) = \lambda \nabla g(x, y)$ and $g(x, y) = 0$